AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 1)

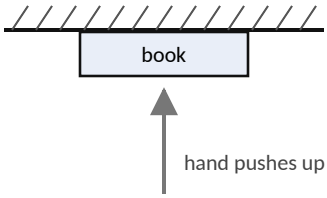
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

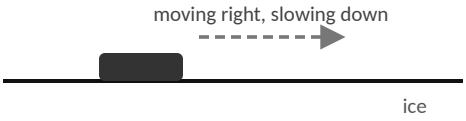
For all questions:

- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) For each situation, draw and label the forces (not components) exerted on the object named. Draw each force as its own arrow starting on the dot and pointing away from it, with lengths that show which forces are larger.



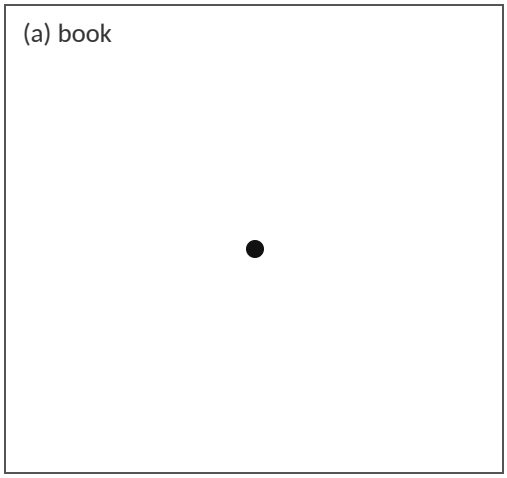
(a)

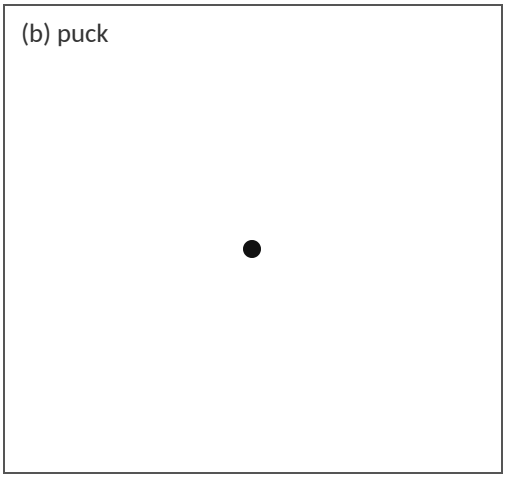


(b)

(a) A hand pushes a 2.0 kg book straight up against the ceiling. The book does not move. Draw the free-body diagram of the book.

(b) A hockey puck slides to the right across the ice and is slowing down. Draw the free-body diagram of the puck.





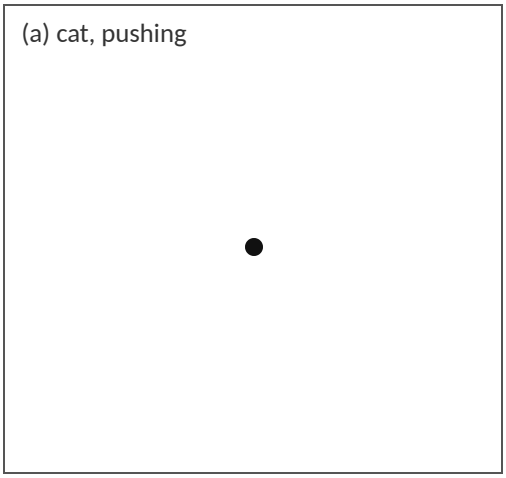
(c) In part (b), which way does the net force on the puck point? Explain how you know without using any numbers.

(d) In part (a), the ceiling pushes down on the book. Identify the Newton's third law partner of that force: what exerts it, on what object, and in which direction.

2. (25 points) A 4.0 kg cat crouches, then pushes straight down on the floor and launches itself straight up. Its paws push on the floor for 0.12 s. After its paws leave the floor, it rises another 0.90 m before it stops at the top. Assume the push is constant.



(a) Draw the free-body diagram of the cat while its paws are pushing on the floor.

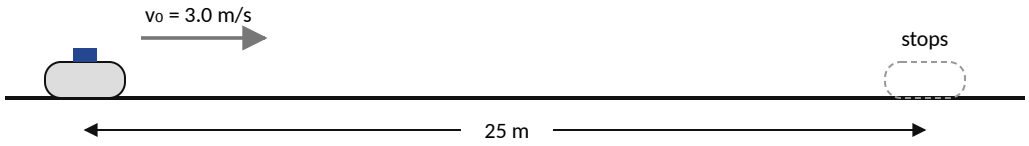


(b) Derive an expression for the magnitude of the force the cat exerts on the floor, in terms of m, g, h and t.

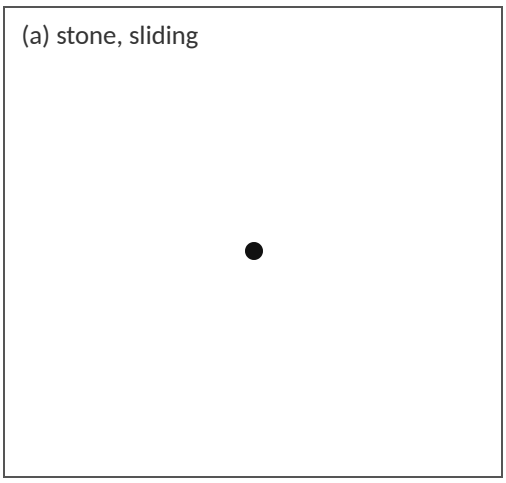
(c) Calculate the magnitude and direction of that force.

(d) How many times the cat's weight is this force?

3. (25 points) A 19 kg curling stone is released at 3.0 m/s on level ice. It slides in a straight line and stops 25 m from where it was released. Friction from the ice is constant and is the only horizontal force on the stone.



(a) Draw the free-body diagram of the stone while it slides.



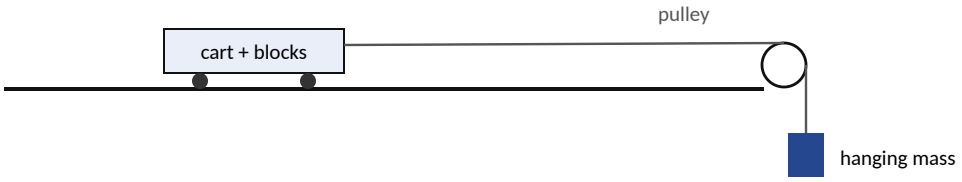
(b) Calculate the stone's acceleration, magnitude and direction.

(c) Calculate the magnitude of the friction force on the stone.

(d) Calculate how long the stone slides.

(e) A second stone has twice the mass, is released at the same speed, and feels twice the friction force. Does it slide farther, a shorter distance, or the same distance? Justify without calculating a number.

4. (25 points) Students pull a cart along a level track with a string that runs over a pulley to a hanging mass. They keep the net force on the cart-and-hanger system the same and change the system mass m by adding blocks to the cart. They measure the acceleration a for each mass.



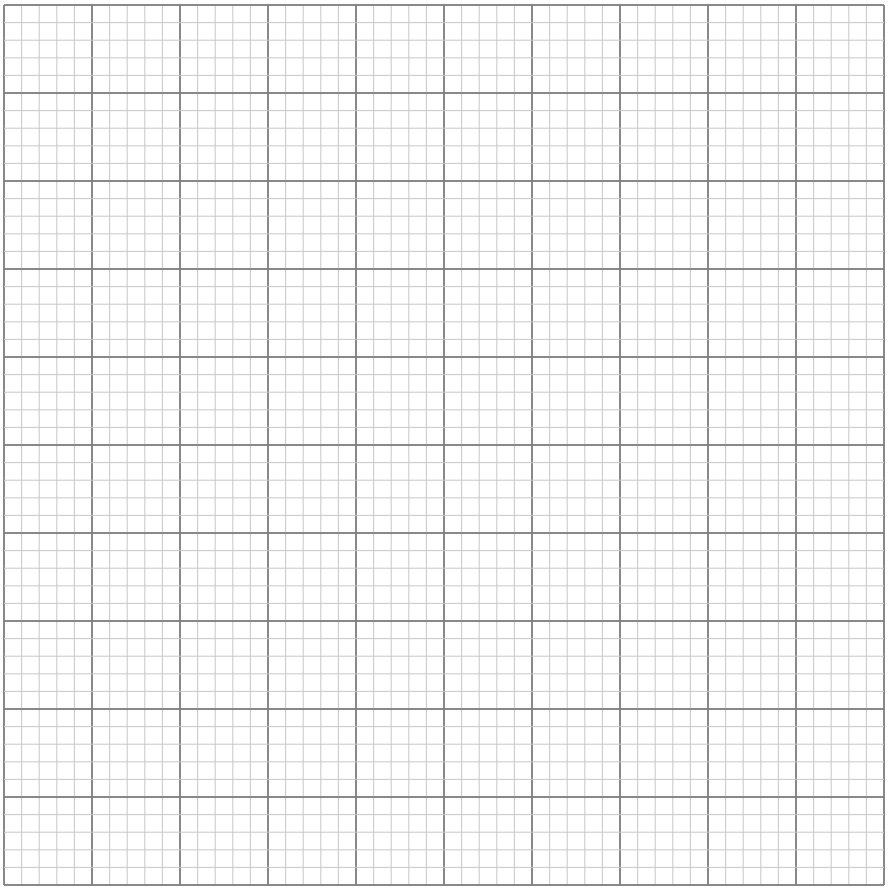
System mass m (kg)	Acceleration a (m/s^2)		
0.250	1.93		
0.350	1.30		
0.450	1.07		
0.550	0.84		
0.650	0.74		
0.750	0.61		

(a) Newton's second law says $a = F_{net}/m$. Choose quantities to graph so the data form a straight line. State what you will plot on the vertical axis and on the horizontal axis, and what the slope of the line represents.

(b) Use the blank column(s) in the table to calculate the values you will plot.

You do not have to use both blank columns. Cross out any column you do not plot.

(c) Plot the data on the grid. Label each axis with the quantity and units, choose a scale, and draw the best-fit line.



(d) Use your line to calculate the net force on the system. Show the two points you use.

(e) The hanging mass is 0.050 kg. Calculate its weight. Then give one physical reason the net force from part (d) could be a little smaller than this weight.

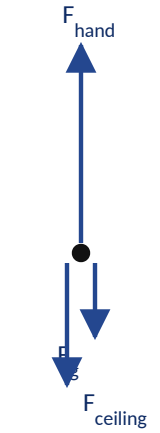
Answer Key – Practice Test, Version 1 – Introductory Dynamics

Worked step by step, the same way as the class notes: sketch, free-body diagram, $a = F_{net}/m$, forces with signs, algebra first, then numbers with units. $g = 9.8\ m/s^2$; answers match the significant figures of the givens.

Every problem: a = ΣF/m (equation sheet), written $a = F_{net}/m$. Replace F_{net} with the actual forces and their signs ($F_N - F_g$, $F_T - F_f$, $F_{th} - F_g$), swap $F_g = mg$, solve in letters, then plug in.
Kinematics, the first three on the sheet: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Free-body diagrams (25 points)

(a) book, at rest



Step 1 **List every force on the book.**

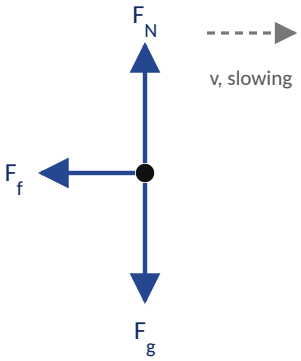
Gravity F_g pulls down. The hand pushes up (F_{hand}). The book is pressed into the ceiling, so the ceiling pushes it down ($F_{ceiling}$). Nothing is under the book, so there is no normal force from below.

Step 2 **Check the lengths.**

At rest means a = 0, so $F_{net} = 0$: $F_{hand} = F_g + F_{ceiling}$. The up arrow is as long as the two down arrows together.

Watch out: the book pushes up on the ceiling, but that force acts on the ceiling, so it does not go on the book's diagram.

(b) puck, sliding right



Step 3 **Vertical, then horizontal.**

No vertical acceleration, so $F_N = F_g$ (equal arrows).

Friction F_f is the only horizontal force. It points opposite the sliding: left.

Watch out: no arrow to the right. Moving right does not need a force to the right; the puck keeps going because of inertia.

Step 4 **(c) Direction of the net force.**

The puck moves right and slows down, so its acceleration points left. $a = F_{net}/m$ and mass is a positive scalar, so F_{net} points the same way as a: **left**

Check: friction is the only horizontal force, and it points left.

Step 5 **(d) Third-law partner.**

$F_{A\text{ on }B} = -F_{B\text{ on }A}$. The partner of $F_{ceiling\text{ on book}}$ is

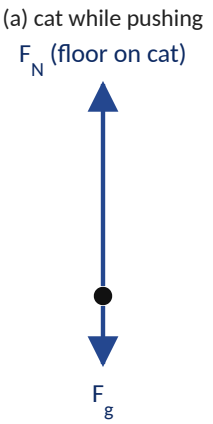
$F_{book\text{ on ceiling}}$: the book pushes up on the ceiling, same magnitude

It acts on a different object (the ceiling), which is why the two never cancel on one diagram.

2. Cat jump: two stages (25 points)

Step 1 **Two accelerations, so two pictures. Up is +.**

stage	known	wanted
1 paws pushing on the floor	$v_0 = 0$, $t = 0.12$ s, $m = 4.0$ kg	a_1, then F_N
2 in the air	v_1 up at liftoff, $v_{top} = 0$, $a = -g$, $\Delta y = h = 0.90$ m	v_1



The cat speeds up upward while its paws push, so the net force is up: the F_N arrow is longer than F_g.

Step 2 **In the air: time is not given or asked, so use #3.**

$v_{top}^2 = v_1^2 + 2a\Delta y \rightarrow 0 = v_1^2 + 2(-g)h \rightarrow v_1 = \sqrt{2gh}$

Step 3 **On the floor: acceleration from the change in velocity.**

$$a_1 = \frac{\Delta v}{\Delta t} = \frac{v_1 - 0}{t} = \frac{\sqrt{2gh}}{t}$$

Step 4 **Newton's second law while pushing. Substitute the expression for a_1, not a number.**

$$a = \frac{F_{net}}{m} \rightarrow a_1 = \frac{F_N - F_g}{m} \rightarrow F_N - mg = ma_1 \rightarrow F_N = m(g + a_1) = m\left(g + \frac{\sqrt{2gh}}{t}\right)$$

Step 5 **Third law: the question asks for the force on the floor.**

$F_{cat\ on\ floor} = F_{floor\ on\ cat} = F_N$, pointing down. **$F = m\left(g + \frac{\sqrt{2gh}}{t}\right)$**

Step 6 **(c) Now the numbers, units in every parenthesis.**

$v_1 = \sqrt{2(9.8\ m/s^2)(0.90\ m)} = \sqrt{17.64\ m^2/s^2} = 4.2$ m/s

$$a_1 = \frac{4.2\ m/s}{0.12\ s} = 35\ m/s^2$$

$F = (4.0\ kg)(9.8\ m/s^2 + 35\ m/s^2) = (4.0\ kg)(44.8\ m/s^2) = 179.2$ N → **1.8×10^2 N, downward on the floor**

Two significant figures, matching 4.0 kg, 0.12 s and 0.90 m.

Step 7 **(d) Compare with the weight.**

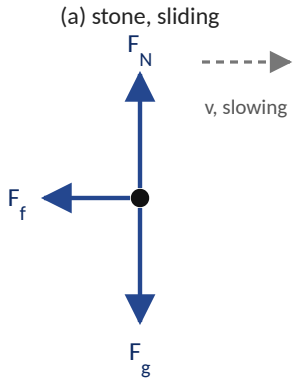
$$\frac{F}{F_g} = \frac{m(g + a_1)}{mg} = 1 + \frac{a_1}{g} = 1 + \frac{35\ m/s^2}{9.8\ m/s^2} = $$ **4.6 times its weight**

Watch out: the 0.12 s is only the time the paws are on the floor. Never use it for the flight, and never use the flight for the push.

3. Curling stone (25 points)

Step 1 **The + direction is the way the stone slides. List the variables.**

v_0	v	a	t	Δx
+3.0 m/s	0 (stops)	?	not given	+25 m



F_N and F_g are equal; friction F_f points backward and is the only horizontal force.

Step 2 **(b) Time is not given and not needed, so use #3.**

$$v^2 = v_0^2 + 2a\Delta x \rightarrow a = \frac{v^2 - v_0^2}{2\Delta x} = \frac{0 - (3.0\ \text{m/s})^2}{2(25\ \text{m})} = \frac{-9.0\ \text{m}^2/\text{s}^2}{50\ \text{m}}$$

$a = -0.18\ \text{m/s}^2$ ($0.18\ \text{m/s}^2$, opposite the motion)

Step 3 **(c) Newton's second law, horizontal.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{-F_f}{m} \rightarrow F_f = -ma = -(19\ \text{kg})(-0.18\ \text{m/s}^2) = 3.42\ \text{N} \rightarrow \boxed{F_f = 3.4\ \text{N}}$$

Step 4 **(d) Time from #1.**

$$v = v_0 + at \rightarrow t = \frac{v - v_0}{a} = \frac{0 - 3.0\ \text{m/s}}{-0.18\ \text{m/s}^2} = 16.7\ \text{s} \rightarrow \boxed{t = 17\ \text{s}}$$

Step 5 **(e) Same distance.**

$a = -\frac{2F_f}{2m} = -\frac{F_f}{m}$: the same acceleration, the same v_0 and the same final v, so $v^2 = v_0^2 + 2a\Delta x$ gives the same Δx. **the same 25 m**

Watch out: no forward force keeps the stone moving. Friction is the only horizontal force, and it points backward.

4. Linearize the data (25 points)

Step 1 **(a) Match $a = F_{net}/m$ to $y = mx + b$.**

$$a = F_{net} \cdot \frac{1}{m} + 0$$

vertical (y)	horizontal (x)	slope	intercept
a (m/s^2)	1/m (kg^{-1})	F_{net}	0

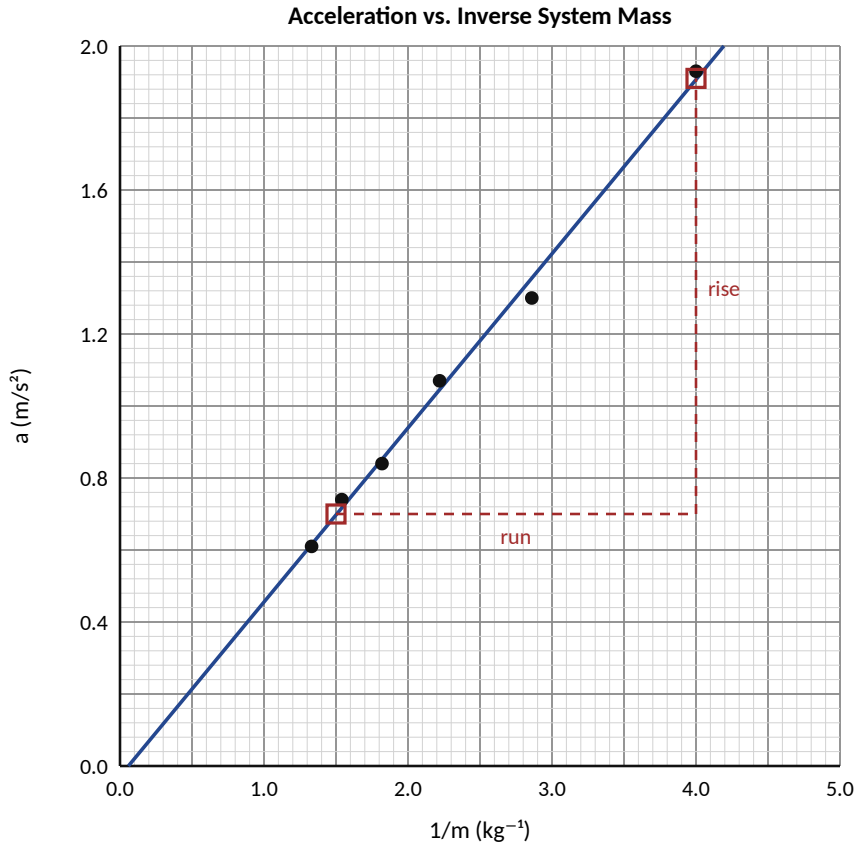
Plot a on the vertical axis and 1/m on the horizontal axis; the slope is F_{net}

Watch out: a vs. m is a curve (a falls off like 1/m), so it is not the line.

Step 2 **(b) Fill in one blank column; cross out the other.**

m (kg)	a (m/s²)	1/m (kg⁻¹)
0.250	1.93	4.00
0.350	1.30	2.86
0.450	1.07	2.22
0.550	0.84	1.82
0.650	0.74	1.54
0.750	0.61	1.33

Step 3 **(c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The line is not forced through the origin. The red squares are points read off the line, not data points.

Step 4 **(d) Slope from two far-apart points on the line, units carried.**

Points on the line: (1.50 kg^{-1}, 0.70 m/s^2) and (4.00 kg^{-1}, 1.91 m/s^2)

$$\text{slope} = \frac{1.91\ m/s^2 - 0.70\ m/s^2}{4.00\ kg^{-1} - 1.50\ kg^{-1}} = \frac{1.21\ m/s^2}{2.50\ kg^{-1}} = 0.484\ kg\cdot m/s^2$$

$\boxed{F_{net} \approx 0.48\ N}$ ($kg\cdot m/s^2 = N$). Any value from 0.44 N to 0.52 N from a reasonable line earns credit.

Step 5 **(e) Weight of the hanger, and why the net force is a little less.**

$F_g = mg = (0.050\ kg)(9.8\ m/s^2) = \boxed{0.49\ N}$

Friction on the cart's wheels and in the pulley pushes against the motion, so the net force on the system is a little less than the hanger's weight.

1. Free-body diagrams

Suggested partial credit	pts
(a) Hand force up, gravity down, ceiling force down, each starting on the dot	5
(a) Up arrow balances the two down arrows; every force labeled	3
(b) Normal force up and gravity down, equal lengths	3
(b) Friction to the left, and no force to the right	4
(c) Left, because the acceleration is left (slowing while moving right)	5
(d) Book on ceiling, upward, same magnitude	5
Total	**25**

2. Cat jump

Suggested partial credit	pts
(a) Normal force up, longer than gravity down; labeled	5
(b) $v_1 = \sqrt{(2gh)}$ from #3 with v = 0 at the top	3
(b) $a_1 = v_1/t$ for the push (not the air time)	2
(b) $F_N - mg = ma_1$ solved for F_N in letters	3
(b) Third-law step: the force on the floor equals F_N	2
(c) Substitution with units in every parenthesis	3
(c) 1.8×10^2 N, downward, two significant figures	3
(d) $1 + a/g = 4.6$	4
Total	**25**

3. Curling stone

Suggested partial credit	pts
(a) Normal up and gravity down, equal; friction backward only	5
(b) Picks #3 because time is not given	2
(b) Solves for a in letters, then substitutes with units	3
(b) -0.18 m/s^2, opposite the motion	2
(c) $-F_f = ma$ set up from $a = F_{net}/m$	3
(c) 3.4 N	2
(d) $t = (v - v_0)/a = 17$ s	4
(e) Same distance: same a = F/m and the same speeds	4
Total	**25**

4. Linearize the data

Suggested partial credit	pts
(a) a on the vertical axis, 1/m on the horizontal axis	3
(a) Slope = F_{net}	3
(b) 1/m column correct	3
(c) Axes labeled with quantity and units	2
(c) Scale uses more than half the grid	2
(c) Points plotted correctly	2
(c) One straight best-fit line, not forced through the origin	2
(d) Two far-apart points on the line; slope with units	3
(d) F_{net} between 0.44 N and 0.52 N	2
(e) 0.49 N, plus friction as the reason	3
Total	**25**